

Qatar-10b

R-0122

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_250629 · created 2026-07-16T05:11:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460855.7446 +/- 0.0091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.086 +/- 0.025
Transit depth (Rp/Rs)^2	0.74 +/- 0.43 [%]
Orbital Inclination (inc)	85.41 +/- 2.41
Airmass coefficient 1 (a1)	1.025 +/- 0.021
Airmass coefficient 2 (a2)	-0.018 +/- 0.018
Scatter in the residuals of the lightcurve fit is	2.06 %
Best Comparison Star	#3 - [345, 271]
Optimal Method	PSF photometry
Transit Duration (day)	0.107 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.086 ± 0.025 vs 0.1265 (deviation 0.69σ)
Duration fit vs published	0.107 d vs 0.1154 d (deviation 0.36σ)
Mid-time O-C	-7.6 min
Depth SNR	1.5
Residual scatter	2.06 %

Light curve

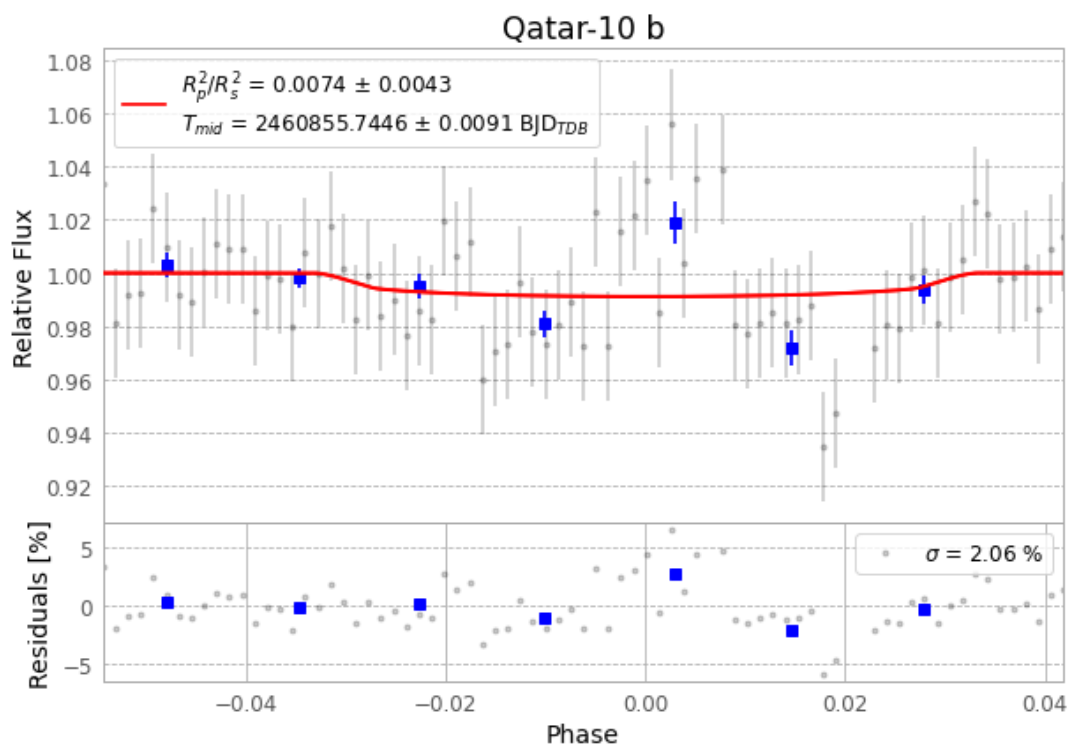


Figure 1 — FinalLightCurve_Qatar-10 b_2025-06-28.png (SHA-256 8830ad548d15624a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [399, 189], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 056f20d2cc272c3f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f0dcec7

Data provenance

77 FITS frames (51 MB), Qatar-10250629034216.FITS → Qatar-10250629073015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-250629.log (bc41a927a56c...), FinalLightCurve_Qatar-10 b_2025-06-28.png (8830ad548d15...), FinalLightCurve_Qatar-10 b_2025-06-28.csv (01db9c142c7a...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:11Z.